

Automatic Video Composition

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Abstract. This paper introduces an approach to generate video clips starting from an annotated commentary. The novelty of the approach lies in the use of rhetorical structure of the accompanying audio commentary in planning the video. The basic notions of cinematography are briefly introduced together with the Rhetorical Relation Theory to model the structure of a discourse. Then, the architecture of a video engine to automatically build video clips from the audio commentary annotated with respect to rhetorical relations is described. Finally, an application for a multimedia mobile guide in a museum is described.

1 Introduction

This paper introduces an approach to generate video clips starting from an annotated commentary and a database of still images. One of the first case studies of the generation of “motion presentations” is the work of (Karp and Feiner, 1993). Their system generates scripts for animation using top down hierarchical planning techniques. (Christianson et al. 1996) presents a successful attempt to encode several of the principles of cinematography in the *Declarative camera control language*.

Similar systems are BETTY (Butz, 94) and CATHI (Butz, 98). BETTY is an animation planner, which generates scripts for animated presentations. The CATHY system generates on-line descriptions of 3D animated clips for the illustration of technical devices, in the context of a coordinated multimedia document.

Animated presentations have been successfully employed also in multimodal frameworks for the generation of explanations (Daniel et al. 1999) and in learning environments (Bares et al. 1997).

A totally different approach is proposed by (Lindley and Nack, 2000). In this work, a planning mechanism is employed to select video clips from a video database and sequencing them into a meaningful presentation for viewers.

The novelty of our approach lies in the use of rhetorical structure of the accompanying audio commentary in planning the video. In particular, knowledge of rhetorical structure is extremely useful in taking decisions related to the punctuation of the video, in order to reflect the rhythm of the audio commentary and its communicative goals. In our view, the verbal part of the documentary always drives the generation of the visual part.

The language of cinematography (Metz, 1974), including shot segmentation, camera movements and transition effects, is employed in order to plan the animation and to synchronize the visual and the verbal parts of the presentation.

Following, we briefly summarize the basic terminology of cinematography and we shortly introduce the Rhetorical Structure Theory to describe the structure of the audio commentary. Then, we will discuss how these conventions can be expressed both in terms of constraints on camera movements and in terms of strategies related to the discourse structure of the associated audio commentary. Finally, a case study of a mobile museum guide that employs automatically composed video clips starting from personalized commentaries will be presented.

2 The Language of Cinematography

According to (Metz, 1974), cinematic representation is not like a human language that is defined by a set of grammatical rules; it is nevertheless guided by a set of generally accepted conventions. These guidelines may be used for developing multimedia presentations that can be best perceived by the viewer.

The shot is the basic unit of a video sequence. In the field of cinematography a shot is defined as a continuous view from single camera without interruption. Since we only deal with still images, we define a shot as a *sequence of camera movements applied to the same image*.

The basic camera movements are *pan*, from “panorama”, a rotation of the camera along the x-axis, *tilt*, a rotation along the y-axis, and *dolly*, a rotation along the z-axis.

Transitions among shots are considered the punctuation symbols of cinematography; they affect the rhythm of the discourse and the message conveyed by the video.

The main transitions used are *cut*, *fade*, and *cross fade*. A cut occurs when the first frame of the following shot immediately replaces the last frame of the current shot. A fade occurs when one shot gradually replaces another one either by disappearing (fade out) or by being replaced by the new shot (fade in). A particular case of a fade happens when instead of two shots, there is one shot and a black screen that can be, again, faded in or faded out. Finally, a cross fade (also called dissolve) occurs when two shots are gradually superimposed during the moment when one is faded out while the other is faded in.

3 Rhetorical Structure Theory

Rhetorical Structure Theory (Mann and Thompson 1987) analyses discourse structure in terms of dependency trees, with each node of the tree being a segment of text. Each branch of the tree represents the relationship between two nodes, where one node is called the nucleus and the other is called the satellite. The information in the satellite relates to that found in the nucleus in that it expresses an idea related to what was said in the nucleus. This rhetorical relation specifies the coherence relation that exists between the two portions of text contained in the nodes. For example, a *Cause*

rhetorical relation holds when the satellite describes the event that caused what is contained in the nucleus. Figure 1, shows an example of a rhetorical tree. Here the second paragraph (the satellite) provides background information with respect to the content expressed in the first paragraph (the nucleus). This additional information acts as a sort of reinforcement for what was previously said in the first paragraph and consequently facilitates the absorption of information. In the original formulation by Mann and Thompson the theory posited twenty different rhetorical relations, while other scholars have since added to this theory.

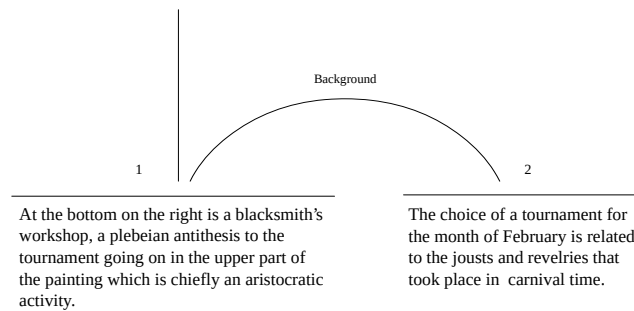


Fig. 1. An example of a rhetorical tree (simplified).

RST was originally developed as part of work carried out in the computer-based text generation field. In a previous work (Not and Zancanaro, 2001), we described a set of techniques to dynamically compose adaptive presentations of artworks from a repository of multimedia data annotated with rhetorical relations. These techniques have been exploited in an audio-based, location-aware adaptive audio guide described in (Not et al., 2000).

4 Video Clips on Still Images

Video clips are built by first searching for the sequence of details mentioned in the audio commentary, deciding the segmentation in shots and then planning the camera movements in order to smoothly focus on each detail in synchrony with the verbal part. In building a video clip, a set of strategies similar to those used in documentaries are employed. Two broad classes of strategies have been identified. The first class encompasses constraints, imposed by the grammar of cinematography, while the second deals with conventions normally used in guiding camera movements in the production of documentaries. While the constraints are just sequence of forbidden camera movements, the conventions are expressed in terms of rhetorical structures found in the audio commentary.

4.1 Constraints on Camera Movements

In order to assure a pleasant presentation, constraints on camera movements have to be imposed. The role of the constraints is to avoid, during the planning of the shots, the creation of a sequence of “discouraged” camera movements, that can lead the user to misunderstand the underlying message carried by the video. For example, a pan from right toward left should not follow a pan from left to right. In general, applying any given movement (pan, tilt and zoom) and then immediately reapplying it on the reverse direction is discouraged because this action renders the video uncomfortable to watch. Figure 2 shows an example of constraint.

```
(defconstraint zoom-in
  (var mv (get-previous-movement))
  (var mv2 (get-previous-movement mv)
    (and
      (not (equal (mv zoom-out)
        (not (equal (mv2 zoom-out))))))
```

Figure 2. An example of constraint

This constraint controls the application of the zoom in movement, which can be used except when the previous or the last but one movement is a zoom out. During the planning of the shots, whenever a camera movement is proposed, all the constraint related to that movements are checked. If they all return true the movement can be applied, otherwise two tricks can be applied: either choosing a different way of focusing the detail required by the verbal part (for example a zoom out can often effectively replace a pan) or starting a new shot altogether. In the latter case, the two shots should be linked by a transition effect that suggests continuity, such as a short fade.

4.2 Rhetorical Strategies

Constraints on camera movements alone are sufficient to ensure a pleasant presentation, yet they do not impact the effectiveness of the video clip. In order to have a more engaging presentation, the visual part should not only focus on the right detail at the right time, but it should also support the presentation of new audio information by illustrating its relation to information that has been already given. In this manner, continuity between the information is built, which in turn facilitates the viewing of the video clip while stimulating the absorption of new information.

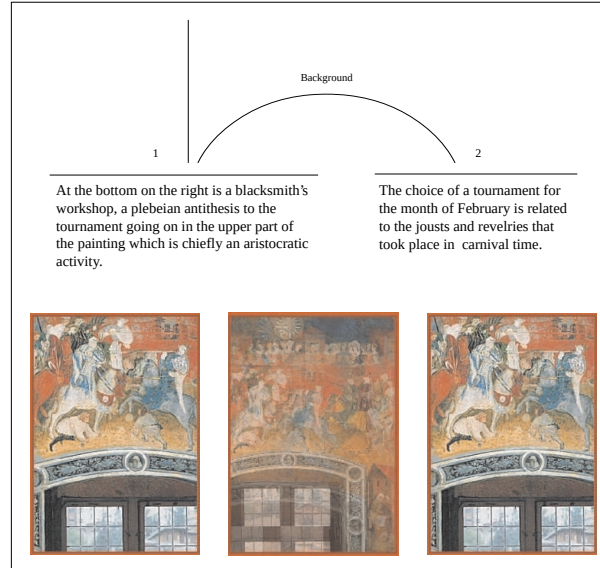


Fig. 3. The “Tournament” example: a cross fade entailed by a background relation

The text in figure 2 can be visually represented with two shots of the same image (that is, the tournament) linked by a long cross fade. Technically, having two shots is not necessary, since the image is the same, but the cross fade helps the user understand background information is going to be provided. The first image is thus presented while the first paragraph is heard over the audio, then when the audio switches to, in this case, the background information, the image is enlarged to cover the entire panel and finally refocused on the detail once the audio has stopped.

A rhetorical strategy suggests, on the basis of a rhetorical tree configuration, what shot segmentation and which transition effect should be applied. The strategies employed in the present prototype were elicited by a focus group activity with a documentary director.

5 Architecture of the Video Engine

The engine is a rule-based system. When a video for a given commentary is requested, the engine analyses the discourse structure of the commentary and selects an appropriate set of images to be presented. The generation chain consists of four phases:

- **Detail Association:** a detail, retrieved from a database of images, is associated with each segment of the commentary;

- **Shot initialization and structure planning:** a candidate structure for the final presentation is elaborated, taking into consideration the rhetorical structure of the commentary;
- **Shot Completion:** camera movements between details are planned. Constraints are considered in order to avoid ``inconsistencies";
- **Editing:** transitions among shots are selected according to the rhetorical structure of the commentary.

The input is an XML representation of the audio commentary. The annotation encompasses the segmentation, the rhetorical relations among segments as well as the duration in seconds and the topic (that is the main concept relative to a simple domain taxonomy) of each segment. The output is the script of the video and the audio channels encoded in a XML.

<pre> <segment id="01" parent="root" relname="none" topic="tournament" audio="castle.wav" duration="3" > At the bottom on the right is a blacksmith's workshop, a plebeian antithesis to the tournament going on in the upper part of the painting which is chiefly an aristocratic activity. </segment> <segment id="02" parent="01" relname="elaboration" topics="castle" audio="windows.wav" duration="2"} /> The differences between the various styles of construction have been reproduced extremely carefully. </segment> </pre>	<pre> <movie id="February"> <shots> ... <play audio="blacksmith.wav"/> <audio-pause duration="1.5"/> </audio-track> </shot> <shot id="shot20645" image="01"> <video-track> <pause duration="7"/> </video-track> <audio-track> <play audio="young-women2.wav"/> </audio-track> </shot> </shots> <editing> <display shot="shot20474"/> <cut duration="1"/> <display shot="shot20480"/> <crossfade shot="shot20522" duration="1.5"/> <crossfade shot="shot20645" duration="1"/> </editing> </movie> </pre>
Fig. 2. Representation of an audio commentary and the corresponding generated script.	

5.1 Phase 1: Detail Association

In this phase the system assigns one or more details to each segment of the commentary. This operation consists in searching a database of images for details with the same topic of the segment. The images in the database were segmented with respect to the details contained and annotated with respect to simple domain taxonomy. The concepts in the taxonomy are the same used in the annotation of the

audio commentary. The result of this phase is the same sequence of segments taken in input such that each segment refers to an image. [Cesare: add example ??]

5.2 Phase 2: Shot initialization

In this phase, shots are initialized taking into consideration the rhetorical structure of the commentary. At the moment the nucleus/satellite distinction is not taken into account. The result of phase 2 is a candidate structure for the final presentation, that is a sequence of shots to be completed. Figure 4 shows an example of shot initialized, which has an id, an image on which camera movements will be applied, a sequence of segments whose content is expressed in the shot and finally an audio- and video-track empty

```
(shot
  id : sh01
  image: "January.jpg"
  segments: (segm01, segm02, segm03)
  audio-track: nil
  video-track: nil)
```

Fig. 4. An example of shot initialized.

The processing is guided by a set of rules, which are fired when particular configurations of rhetorical relations are matched. For example a relation of type elaboration or sequence signals a smooth transition from the current topic to new information that is strictly related to it; it is thus preferable to aggregate segments in the same shot and to exploit camera movements. Background and circumstance tend to highlight the introduction of new information that provides a context in which the following or the previous messages can be interpreted. They tend to break the flow of the discourse. It is thus preferable to split the segments in two different shots so that, in the next phase, it is possible to exploit proper transition effects in order to emphasize that change of rhythm.

There are cases in which the structure planned in this phase is revised during successive stages of computation. For example, to avoid the "seasickness" effect the system can apply constraints and then modify the previously planned structure by adding new shots.

5.3 Phase 3: Shot completion

This is the phase in which the engine incrementally completes each shot by illustrating each of its segments. In performing this task the engine traces the camera movements already planned. When a candidate move is proposed the system verifies whether it is suitable or not according to the list of past camera movements and the constraints imposed over that type of movement. Figure 5 shows a completed shot, where the list of the segments is empty whereas audio- and video-track contain a sequence of action applied to images or audio files.

```

(shot
  id : sh01
  image: "January.jpg"
  segments: empty
  audio-track: ((play :sound "january.wav")
                (a-pause :duration 2))
  video-track: ((pause :duration 2)
                (zoom :scale 1.20 :duration 3)))

```

Fig. 5. An example of shot completed.

Constraints encode the cinematographer's expertise in selecting and applying camera movements in order to obtain "well-formed" shots. For instance, when a panning movement is proposed where the previous movement is also a panning, the system has to check if the resulting sequence is suitable. Simple constraints include: (i) When the previous movement is a dolly-out a dolly-in cannot be applied; (ii) When the previous movement is a dolly-in a dolly-out cannot be the subsequent movement; (iii) When a panning or a tilting is along a similar path and in the opposite direction of the previous movement that panning or tilting cannot be applied.

Constraints encode *schemes* of forbidden movements and when one of them is not satisfied the proposed move is rejected. In this case the engine initializes a new shot, declares the previous one completed and associates the remaining segments to the new shot.

5.4 Phase 4: Movie Editing

This is the phase in which the engine chooses the "punctuation" of the presentation. Movie editing is achieved by selecting appropriate transitions among shots. In order to reflect the rhythm of the discourse, the choice of transition effects is guided by the rhetorical structure of the commentary. The system retrieves the last segment of the shot displayed and the first segment of the shot to be presented and plans the transition according to the following rules:

- If two segments are linked by a relation of type elaboration a short cross fade applies;
- If two segments are linked by a relation of type background or circumstance a long cross fade applies.
- If two segments are linked by a relation of type sequence a cut applies.
- If a relation of type enumeration holds among two or more segments a rapid sequence of cut applies.

These rules have been selected according to the observations about the usual employment of transition effects in the field of cinematography (Arijon, 1976) and from focus group with a documentary director. Fade effects are fit for smooth transition, when there is a topic shift or when the center of interest changes but the new topic is related to the old one, as in the case of elaboration or background. Cut is more appropriate for abrupt and rapid changes, to emphasize the introduction of a new concept, as in the case of sequence. A special case holds when the verbal commentary enumerates a set of subjects or different aspects of the same object; in

those cases a rapid sequence of cuts can be used to visually enumerate the elements described. The output of this phase is a complete script with the descriptions of shots and the sequence of transition effects to be applied to shots (editing).

6 Cinematography for a Mobile Multimedia Guide

As a test case of the generation of video clips, we are investigating the use of video clips in a multimedia mobile guide in a museum. Many research projects are exploring the new possibilities offered by Personal Digital Assistants (PDAs) in a museum setting (see for example, Grinter et al, 2002 and Cheverst 2000). Usually, these multimedia guides use static images, while others employ pre-recorded short video clips about museum exhibits. Our hypothesis is that using video clips to present the description of a painting allows the visitor to better identify the details introduced by the audio counterpart of the presentation. Moreover, coupling the video clip engine with a module that produces verbal description of tailored information (such the one described in Not and Zancanaro, 2001), the guide can provide adaptive video documentaries,

At present, we have completed a first prototype for the famous XV century fresco “The Cycle of the Months” at Torre Aquila in Trento, Italy. It illustrates the activities of aristocrats and peasants throughout the year and covers all four walls of a tower. It introduces a number of characters as well as many different activities from falconry to wine harvesting.

The guide is implemented on a PDA, which by means of infrared sensors is capable of identifying its position within the frescoed Tower of the castle. Interaction with the system is both proposed by the system itself and accepted by the user, thus sharing the responsibility of information access. When the system detects that the visitor is in front of one of the four walls, a picture of that wall is displayed on the PDA and, after a few seconds, if the user has not changed her position the panel she is facing is highlighted (see figure 4). At this point, the visitor can click on the panel and receive a multimedia presentation of the panel she has chosen. This modality has been chosen in order to allow the visitor to retain control of an inherently proactive guide.



Fig. 5. Snapshots the Multimedia Guide

Preliminary studies and pilot tests show encouraging results and interesting effects. All users became acquainted with the system very quickly. Most of them used the PDA as a “3D mouse”, pointing directly to the infrared emitters to speed up the localization. In the future, we will propose a new interface where the user can be explicitly involved in the process of localization. Most of the users complained before actually using the system that a video sequence on a PDA would distract their attention from the real artwork. After a short interaction with the system, however, they appreciated the possibility of quickly localizing small details on the fresco. This demonstrates that use of cinematic techniques in a multimedia guide can be effective, particularly when explaining a complex painting.

7 Conclusions and Future Work

The paper describes an engine to compose video clips starting from an annotated audio commentary and a repository of images. An application for a multimedia mobile guide in a museum has been introduced. The novelty of the approach lies in the idea of using the Rhetorical Structure Theory to describe the semantics of the transitions among shots.

To improve the quality of the resulting presentation we are planning to introduce new rules in order to allow: (i) the use of composed camera movements, (ii) the variation of the speed of the camera and (iii) the use of “metagraphics” (arrows, bounding boxes, blurred areas) as an alternative to zoom in, in order to highlight also the very detailed parts of the images.

At the moment the application does not take into account user-dependent features like age, preferences, level of expertise, etc. We are planning to develop a new version of the software that can also allow the variation of the rhythm of the documentary according to the needs of a particular user.

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